Week 11 — Deployment Report

Deployment milestone

CSE4204-8B-T04 | Northern University of Business and Technology, Khulna

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1. Deployment Objective

This report documents the Week 11 deployment milestone: making the completed Enterprise NeXus system available online through independent frontend, backend, database, and automation services. The deployment keeps secrets server-side, supports redeployment from GitHub, and exposes health endpoints for monitoring.

2. Team Information

Field	Detail
Course	CSE 4204 — Mobile Computing Lab
Team	CSE4204-8B-T04
Project	Enterprise NeXus — Multi-Agent AI Business Automation OS with Security Intelligence
Supervisor	Md. Riaz Mahmud (Assistant Professor)
Institution	Northern University of Business and Technology, Khulna

Name	Student ID	Role
Nazmus Sakib	11220320888	Team Leader & AI Integration Lead
Shoeb Shariar Mashuk	11220320878	Backend Developer
Most Sumiya Sanjida	11220320874	Frontend Developer
Sabrina Ibrahim	11220320895	UI & Frontend Developer

3. Production Architecture

Layer	Platform	Responsibility
Frontend	Vercel	Next.js dashboard, store, careers portal, responsive UI, and API rewrite
Backend API	Render	Node.js/Express REST API, authentication, AI agents, webhooks, and health checks
Database and retrieval	Supabase	PostgreSQL, Row-Level Security, realtime data, and pgvector search
Automation	n8n	Scheduled reports, retries, callbacks, and operational workflows
Monitoring	Render, UptimeRobot, Sentry	Availability, errors, and incident visibility

4. Live Deployment Links

- Main application: https://enterprise-nexus-delta.vercel.app
- Customer store: https://enterprise-nexus-delta.vercel.app/store
- Careers portal: https://enterprise-nexus-delta.vercel.app/careers
- Backend health: https://enterprise-nexus-backend.onrender.com/api/health
- Backend readiness: https://enterprise-nexus-backend.onrender.com/api/ready
- Repository: https://github.com/Nazmussakib247/CSE4204-8B-T04-Enterprise-Nexus-Multi-Agent-AI-Business-Automation-OS-with-Security-Intelligence

5. Deployment Procedure

1. Push the reviewed release state to GitHub.
2. Deploy the backend from backend/ on Render using render.yaml and Dockerfile.
3. Configure Render environment variables without committing secret values.
4. Deploy the frontend from frontend/ on Vercel and set public API variables.
5. Set FRONTEND_URL on Render to the deployed Vercel origin for CORS.
6. Apply Supabase migrations in filename order and verify Row-Level Security.
7. Import n8n workflows and configure N8N_SECRET for callbacks.

8. Run the post-deployment verification checklist before sharing links.

6. Environment Variables and Security

API keys, database credentials, and JWT secrets are never committed or shipped to the browser. Local values are stored in ignored .env files; production values are configured in the service dashboards.

- Backend: SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, SUPABASE_ANON_KEY, JWT_SECRET, GEMINI_API_KEY, FRONTEND_URL, N8N_WEBHOOK_URL, and N8N_SECRET.
- Frontend: NEXT_PUBLIC_API_URL, NEXT_PUBLIC_SUPABASE_URL, and NEXT_PUBLIC_SUPABASE_ANON_KEY.
- Helmet, CORS, rate limiting, validation, audit logs, correlation IDs, JWT revocation, and Row-Level Security protect production operations.
- Health and readiness endpoints bypass application rate limiting so UptimeRobot does not create false 429 failures.

7. Post-Deployment Verification

Check	Expected Result	Status
Frontend	Application and public navigation load	Verified
Backend health	GET /api/health returns a healthy status	Verified
Authentication	Registration, sign-in, refresh, logout, and reset work	Verified
Core modules	HR, Finance, Support, Analytics, and Executive load	Verified
AI reliability	Validation, retry, fallback, and escalation paths available	Verified
Monitoring	Health endpoint configured for uptime monitoring	Configured

8. Operational Notes

- Render free-tier services may sleep after inactivity; the first request can be slower during wake-up.
- UptimeRobot should monitor the backend health endpoint rather than a protected application route.
- A failed release should be rolled back to the last known-good GitHub commit.
- Production data, credentials, and uploads must follow the project environment policy.

9. Deliverables Summary

The Week 11 deployment milestone is complete. Enterprise NeXus has a live Vercel frontend, Render backend, Supabase database and vector retrieval, n8n automation support, environment-based configuration, health monitoring, and documented verification steps.